Sigma Notation: Index Shifts and Convergence Conditions Answer Key

Precalculus

Quick Answers

1. 35
2. 16
3. 91
4. 5
5. 48.75
6. 0
7. $\frac{4}{3}$
8. 2025

Common wrong answers

1. *If they got 36: started the index at zero instead of one, adding one extra term*
2. *If they got 15: subtracted the limits without adding one, the fencepost error*
3. *If they got 140: shifted the summand but left the upper limit unchanged, adding one extra term*
4. *If they got 2: used the n equals one term as the first term although the sum starts at n equals zero*
7. *If they got 3: used one as the first term, as if the sum began at n equals zero*
8. *If they got 6084: kept the original upper limit after shifting the summand, adding three extra cubes*

Two habits this sheet builds. (1) A sigma sum is the summand *and* its limits: change one, change the other, and check that the number of terms survived. (2) The geometric sum formula is a conditional statement — name a and r, verify $|r| < 1$, and only then divide.

1. Evaluate $\sum_{n=1}^{5}(2n+1)$.

The index runs over the integers from 1 to 5, so there are $5 - 1 + 1 = 5$ terms. Write them out by substituting each value of n into $2n + 1$:

$$3 + 5 + 7 + 9 + 11$$

Pairing the outside terms, $3 + 11 = 14$ and $5 + 9 = 14$, leaves 7 in the middle: $14 + 14 + 7$. The sum is $\boxed{35}$.
Watch for: 36, which comes from starting at $n = 0$ and picking up the extra term 1. The lower limit is part of the definition, not a formality.

2. How many terms are in $\sum_{k=4}^{19} a_k$?

Counting the integers from 4 to 19 inclusive is a fencepost count: subtracting gives the number of *gaps*, and there is always one more post than gap. So add one after subtracting:

$$19 - 4 + 1$$

The sum has $\boxed{16 \text{ terms}}$.
Watch for: 15, the difference alone. A quick sanity check on a small case settles it: from 4 to 5 there are clearly 2 terms, not 1.

3. Find the mistake: Ravi's shift of $\sum_{n=2}^{7}(n-1)^2$.

(a) Ravi changed the summand but left the limits alone. Substituting $m = n-1$ shifts *every* index value down by one, so the limits must move too: when $n = 2$, $m = 1$; when $n = 7$, $m = 6$. His version runs m from 1 to 7, which is 7 terms where the original has only 6 — he silently added a term.
(b) The correctly shifted sum is

$$\sum_{n=2}^{7}(n-1)^2 \;=\; \sum_{m=1}^{6} m^2 \;=\; 1+4+9+16+25+36$$

which totals $\boxed{91}$.
Check the count first: original $7-2+1=6$ terms, shifted $6-1+1=6$ terms. Equal counts is the signal that the shift was done correctly, and it is what rules out Ravi's 140.

4. Evaluate $\sum_{n=0}^{\infty} 3\left(\frac{2}{5}\right)^n$.

Name the pieces before using any formula. The sum starts at $n=0$, so the first term is $a = 3\left(\frac{2}{5}\right)^0 = 3$, and the ratio is $r = \frac{2}{5}$. Since $\left|\frac{2}{5}\right| < 1$, the series converges and $\frac{a}{1-r}$ applies:

$$\frac{3}{1-\frac{2}{5}} = \frac{3}{\frac{3}{5}}$$

The sum is $\boxed{5}$.
Watch for: 2, which comes from using the $n=1$ term $\frac{6}{5}$ as a. The first term is whatever the *stated* lower limit produces.

5. Find the mistake: Tess and $\sum_{n=1}^{\infty} 4\left(\frac{3}{2}\right)^n$.

(a) Tess never checked the convergence condition. The formula $\frac{a}{1-r}$ is valid only when $|r| < 1$; here $r = \frac{3}{2}$ and $\left|\frac{3}{2}\right| > 1$, so the formula does not apply. Her answer also fails an obvious sanity test: every term is positive, so no partial sum can be negative, let alone the total.
(b) The terms are 6, 9, 13.5, 20.25, and adding the first four gives

$$6+9+13.5+20.25 = \boxed{48.75}$$

Each term is larger than the one before, so the partial sums are not settling toward anything — they grow without bound.
(c) The series **diverges**: it has no sum. "Diverges" is the complete answer here, and -8 is not a wrong sum so much as a number produced by a formula that was never licensed to run.

6. (a) $\lim_{n\to\infty} \frac{1}{n}$, and (b) what it does not prove.

(a) As n grows without bound the denominator grows without bound while the numerator stays fixed, so the quotient can be made smaller than any positive number: $\boxed{0}$.
(b) Part (a) is a statement about the *terms*, and convergence of a series is a statement about the *partial sums*. Terms shrinking to 0 is necessary but not sufficient: the harmonic series $\sum \frac{1}{n}$ has terms tending to 0 and still diverges, because the terms do not shrink fast enough (grouping them into blocks of $2, 4, 8, \ldots$ terms gives blocks each summing to at least $\frac{1}{2}$).
So the nth-term test can only ever prove *divergence* — when the terms do *not* tend to 0. When they do tend to 0, the test is silent and another test is needed.

7. Evaluate $\sum_{n=2}^{\infty}\left(\frac{2}{3}\right)^n$.
The ratio is $r = \frac{2}{3}$ and $\left|\frac{2}{3}\right| < 1$, so the series converges. The trap is a: the sum starts at $n = 2$, so the first term is $\left(\frac{2}{3}\right)^2 = \frac{4}{9}$, not 1.

$$\frac{a}{1-r} = \frac{\frac{4}{9}}{1-\frac{2}{3}} = \frac{4}{9}\cdot\frac{3}{1}$$

The sum is $\boxed{\dfrac{4}{3}}$.
Watch for: 3, the answer to the sum that starts at $n = 0$. The formula's a is always the first term actually included.

8. Rewrite $\sum_{n=4}^{12}(n-3)^3$ starting at 1, then evaluate.
Substitute $m = n - 3$. Then $n = 4$ gives $m = 1$ and $n = 12$ gives $m = 9$, and the summand $(n-3)^3$ becomes m^3:

$$\sum_{n=4}^{12}(n-3)^3 = \sum_{m=1}^{9} m^3$$

Term counts agree: $12 - 4 + 1 = 9$ and $9 - 1 + 1 = 9$. Now use the sum-of-cubes identity, which is the square of the corresponding triangular number:

$$\sum_{m=1}^{9} m^3 = \left(\frac{9\cdot 10}{2}\right)^2 = 45^2$$

The sum is $\boxed{2025}$.
Watch for: 6084, which is $\sum_{m=1}^{12} m^3$ — the summand shifted while the upper limit stayed behind, the same error as problem 3 at a larger scale.